TWO STAGE REDUCER GEARBOX

SUMMER INTERNSHIP PROJECT REPORT

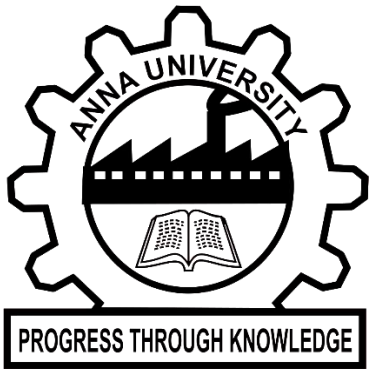
Submitted by

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SCoE-MIT Summer Internship - 2026

On

CAD AND CAE



SIEMENS CENTRE OF EXCELLENCE (SCoE)

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10.07.2026

BONAFIDE CERTIFICATE

Certified that the SCoE-MIT Summer Internship Project Report entitled "TWO STAGE REDUCER GEARBOX "is the Bonafide work of **RAMCHAND R N 950023114038 AURCT**

who carried out the project work. Certified further that, to the best of my knowledge, the work reported here does not form part of any project on the basis of which degree or award was conferred on an earlier occasion on this or any other candidate.

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RAMCHAND R N

ABSTRACT

A two-stage reduction gearbox is a mechanical power transmission system designed to reduce the rotational speed of the input shaft while increasing the output torque. This project focuses on the **design, modelling, and assembly of a two-stage reduction gearbox using Siemens NX CAD**. The gearbox consists of essential mechanical components such as spur gears, input, intermediate, and output shafts, bearings, retaining rings, and a split gearbox housing. The use of two gear reduction stages enables a higher overall reduction ratio, ensuring efficient power transmission with improved torque output and smooth operation.

Each component of the gearbox was created as an individual 3D model in **NX CAD** using parametric design techniques and assembled using appropriate assembly constraints to achieve accurate positioning and alignment. Standard machine elements, including bearings and shafts, were incorporated to ensure proper support, reduce friction, and maintain rotational stability. The gearbox housing was designed to protect the internal components while allowing ease of assembly, inspection, and maintenance. Special attention was given to gear meshing, shaft alignment, and bearing placement to ensure reliable performance.

The assembled model demonstrates the working principle of a two-stage reduction gearbox, where power is transmitted from the input shaft to the intermediate shaft through the first gear pair and then to the output shaft through the second gear pair. This arrangement significantly reduces the output speed while increasing torque, making the gearbox suitable for industrial applications such as conveyors, machine tools, automation equipment, material handling systems, and other power transmission mechanisms.

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CHAPTER 1 – INTRODUCTION

A gearbox is one of the most important mechanical power transmission systems used in various industrial and automotive applications. Its primary function is to transmit power from a driving shaft to a driven shaft while modifying the speed and torque according to the application requirements. By reducing the rotational speed of the input shaft, a gearbox increases the torque available at the output shaft, enabling machines to perform heavy-duty operations efficiently. Among the various types of gearboxes, the **two-stage reduction gearbox** is widely used because it provides a higher speed reduction ratio while maintaining compact size, smooth operation, and improved efficiency.

A two-stage reduction gearbox consists of two pairs of meshing gears mounted on three shafts namely the input shaft, intermediate shaft, and output shaft. In the first stage, power is transmitted from the input shaft to the intermediate shaft through the first pair of spur gears. The intermediate shaft then transfers the motion to the output shaft through the second pair of gears, thereby achieving an additional reduction in speed. This two-stage arrangement allows a large reduction ratio to be obtained without using excessively large gears, making the gearbox suitable for applications requiring high torque and controlled rotational speed.

With the rapid advancement of Computer-Aided Design (CAD) technology, the design and development of mechanical systems have become more accurate, efficient, and economical. **Siemens NX CAD** is one of the most advanced CAD/CAM/CAE software packages used in industries for product design, assembly modelling, simulation, and manufacturing. It enables engineers to create detailed three-dimensional models, perform precise assemblies, detect interferences between components, and modify designs easily using parametric modelling techniques. The software significantly reduces design time while improving product quality and accuracy.

In this project, a complete **two-stage reduction gearbox** has been designed and assembled using **Siemens NX CAD**. The gearbox consists of major mechanical components such as spur gears, shafts, bearings, retaining rings, keys, and a split gearbox housing. Each component was modelled individually with appropriate

arrangement, shaft positions, bearing locations, and housing structure to achieve a reliable and functional assembly.

The assembly model demonstrates the working principle of the gearbox by illustrating the transmission of power from the input shaft to the output shaft through two stages of gear reduction. Proper gear meshing, shaft alignment, and bearing support ensure smooth power transmission and minimize vibration and friction. The split housing design allows convenient assembly, inspection, lubrication, and maintenance of the gearbox components.

Two-stage reduction gearboxes are extensively used in industrial machinery such as conveyors, rolling mills, machine tools, mixers, crushers, hoists, elevators, packaging equipment, and automated manufacturing systems. Their ability to provide high torque with controlled speed makes them indispensable in modern mechanical engineering applications. The use of spur gears in this project offers advantages such as simple construction, high efficiency, ease of manufacturing, and reliable operation under moderate load conditions.

The primary objective of this project is to gain practical knowledge in mechanical design, assembly modelling, and product development using **Siemens NX CAD**. It provides an understanding of gearbox construction, gear train arrangement, shaft design, bearing selection, and assembly techniques while improving proficiency in CAD modelling. The project also demonstrates the importance of dimensional accuracy, component compatibility, and proper assembly constraints in designing a functional mechanical system.

Overall, the design and assembly of the two-stage reduction gearbox using **Siemens NX CAD** serves as an effective engineering project that combines theoretical knowledge with practical design skills. It enhances understanding of mechanical power transmission systems while developing the technical competencies required for industrial product design and manufacturing. This project also provides a strong foundation for future work involving gearbox analysis, optimization, finite element analysis (FEA), motion simulation, and manufacturing process planning.

CHAPTER 2 – OBJECTIVE

The primary objective of this project is to design and assemble a **Two-Stage Reduction Gearbox** using **Siemens NX CAD**, providing a comprehensive understanding of mechanical power transmission systems and modern computer-aided design techniques. The project aims to develop a complete three-dimensional model of the gearbox by creating individual components such as spur gears, input shaft, intermediate shaft, output shaft, bearings, keys, retaining rings, and the gearbox housing with accurate dimensions and engineering standards. Each component is designed separately and then assembled using appropriate assembly constraints to ensure proper positioning, alignment, and smooth interaction between all parts. The project emphasizes the importance of precision in mechanical design, ensuring that every component functions correctly within the complete assembly.

Another important objective is to understand the working principle of a two-stage reduction gearbox and the mechanism of transmitting power from the input shaft to the output shaft through two successive stages of gear reduction. The gearbox is designed to reduce the rotational speed while increasing the output torque, making it suitable for industrial machinery that requires controlled speed and high-power transmission. Proper gear meshing, shaft arrangement, bearing placement, and housing design are carefully considered to ensure efficient operation, reduced friction, and reliable performance. The project also aims to study the relationship between gear ratios, speed reduction, and torque multiplication, which are fundamental concepts in machine design and mechanical engineering.

This project further aims to enhance practical skills in **Siemens NX CAD** by applying parametric modelling, part modelling, assembly design, and engineering drawing techniques. Through this process, the user gains hands-on experience in designing complex mechanical assemblies while understanding the importance of dimensional accuracy, geometric constraints, and component compatibility. The assembly process also helps in identifying possible interference between components

In addition, the project seeks to bridge the gap between theoretical knowledge and industrial practice by demonstrating how CAD software is used in the product development process. It provides exposure to standard mechanical components, gearbox construction methods, and assembly procedures that are widely followed in manufacturing industries. The completed gearbox model serves as a foundation for future engineering activities such as motion simulation, finite element analysis (FEA), stress analysis, manufacturing process planning, and design optimization. Overall, the objective of this project is not only to develop a functional two-stage reduction gearbox but also to strengthen engineering design skills, improve proficiency in Siemens NX CAD, and gain practical knowledge of mechanical power transmission systems used in real-world industrial applications.

CHAPTER 3 – SOFTWARE TOOL

NX CAD: Empowering Design Innovation

NX CAD, developed by Siemens Digital Industries Software, stands at the forefront of computer-aided design (CAD) software, renowned for its robust capabilities and comprehensive feature set. This essay explores the key features and benefits of NX CAD, highlighting its significance in modern design processes. CAD is technology concerned with using computer systems to assist in the creation, modification, analysis, and optimization of a design. Any computer program that embodies computer graphics and an application program facilitating engineering functions in design process can be classified as CAD software.

An NX 12 sketch is a named set of curves joined in a string that when swept, form a solid. The sketch represents the outer boundary of that part. The curves are created on a plane in the sketcher. In the beginning, these curves are drawn without any exact dimensions. Then, Dimensional Constraints as well as Geometric Constraints are applied to fully constrain the sketch

Comprehensive Design Capabilities

NX CAD offers a vast array of tools tailored to meet the complex demands of product design. At its core, the software enables engineers and designers to create4 precise 3D models with unparalleled accuracy. The parametric modelling feature allows for the creation of intelligent models where design intent can be easily biomodified and updated, facilitating iterative design processes without compromising integrity.

Integrated Design Environment

One of NX CAD's strengths lies in its integrated environment, which seamlessly combines various design disciplines such as mechanical, electrical, and industrial design. This integration fosters interdisciplinary collaboration, enabling teams concurrently on different aspects of a project within a unified platform. Moreover, NX CAD supports data interoperability with other CAD systems. Facilitating seamless data exchange and collaboration across the supply chain.

Advanced Simulation and Analysis

Beyond basic modelling, NX CAD incorporates advanced simulation and analysis tools that empower engineers to validate designs virtually. Finite Element Analysis (FEA), Computational Fluid Dynamics (CFD), and motion simulation capabilities allow for comprehensive performance assessments early in the design phase. This iterative simulation-driven design approach helps identify potential issues, optimize designs for performance and efficiency, and ultimately reduce costly physical prototypes.

Digital Twin and Lifecycle Management

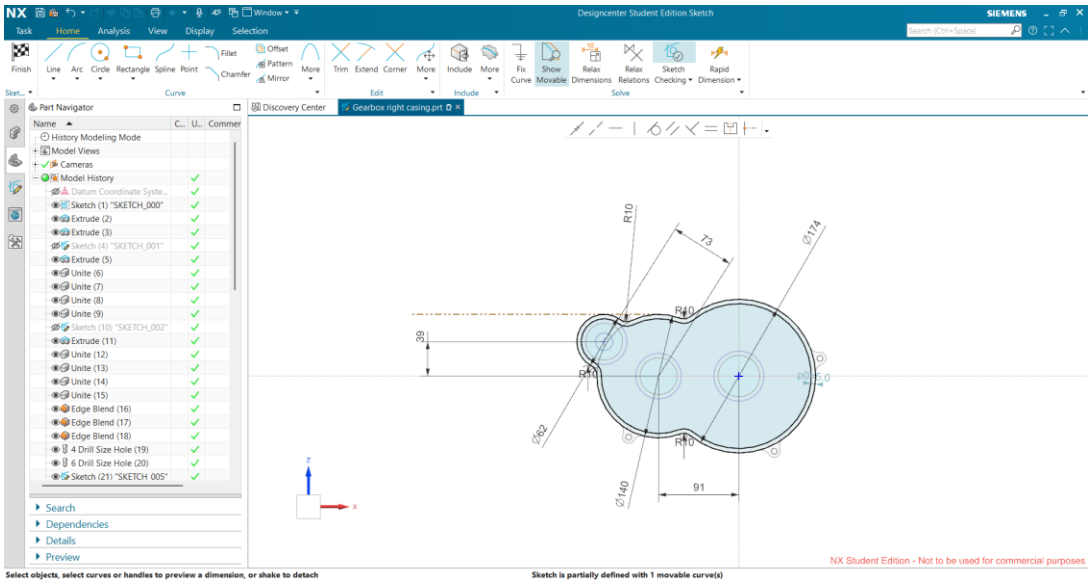
NX CAD supports the concept of the digital twin, where a virtual representation of the product is synchronized with its physical counterpart throughout its lifecycle. This digital continuity enables real-time monitoring, analysis, and optimization of product performance, from initial design through manufacturing, operation, and maintenance phases. Integrated lifecycle management features ensure that design changes are efficiently tracked, documented, and communicated across stakeholders.

Manufacturing Integration

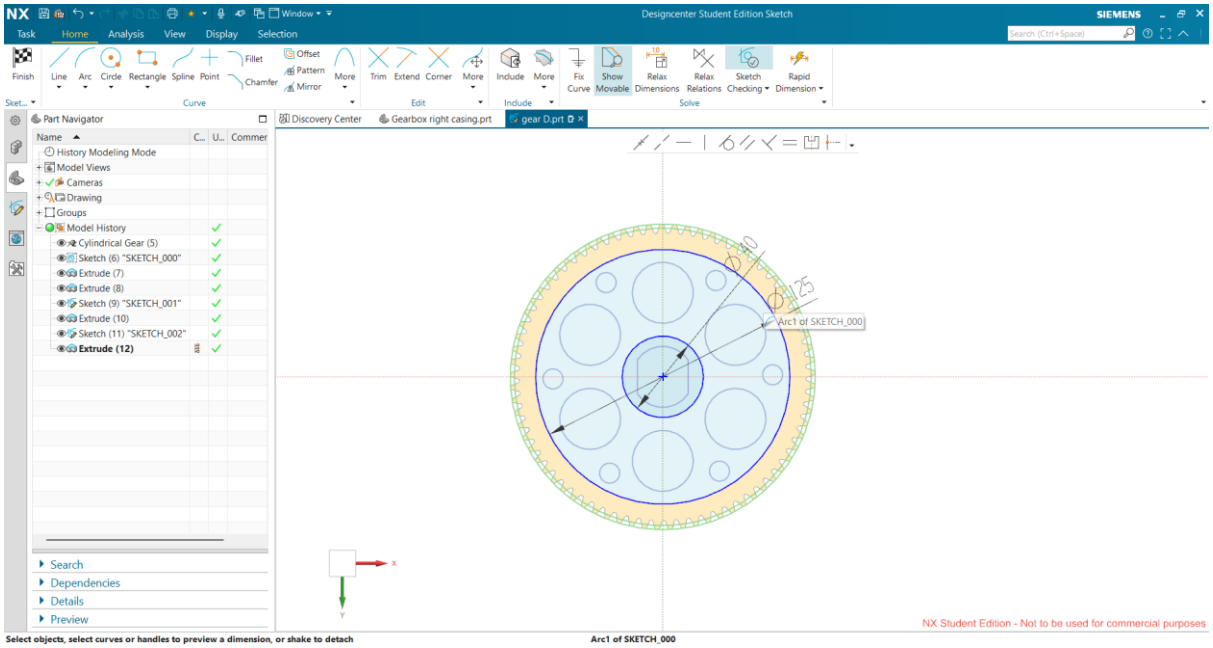
For manufacturing readiness, NX CAD offers robust tools for tooling and NC programming. Its integrated CAM functionality enables the generation of efficient tool paths for machining operations, ensuring accuracy and minimizing production lead times. Additionally, NX CAD supports additive manufacturing processes, enabling designers to optimize designs for 3D printing and other additive techniques. In conclusion, NX CAD represents a pinnacle of innovation in CAD software, empowering designers and engineers with advanced tools for creating, simulating, and managing complex product designs. Its comprehensive feature set, integrated environment, and support for digital twin and lifecycle management make it indispensable in industries ranging from automotive and aerospace to consumer goods and electronics. By leveraging NX CAD's capabilities, organizations can accelerate innovation, improve product quality, and maintain a competitive edge in today’s rapidly evolving global marketplace.

CHAPTER 4 – DESIGINING STEPS

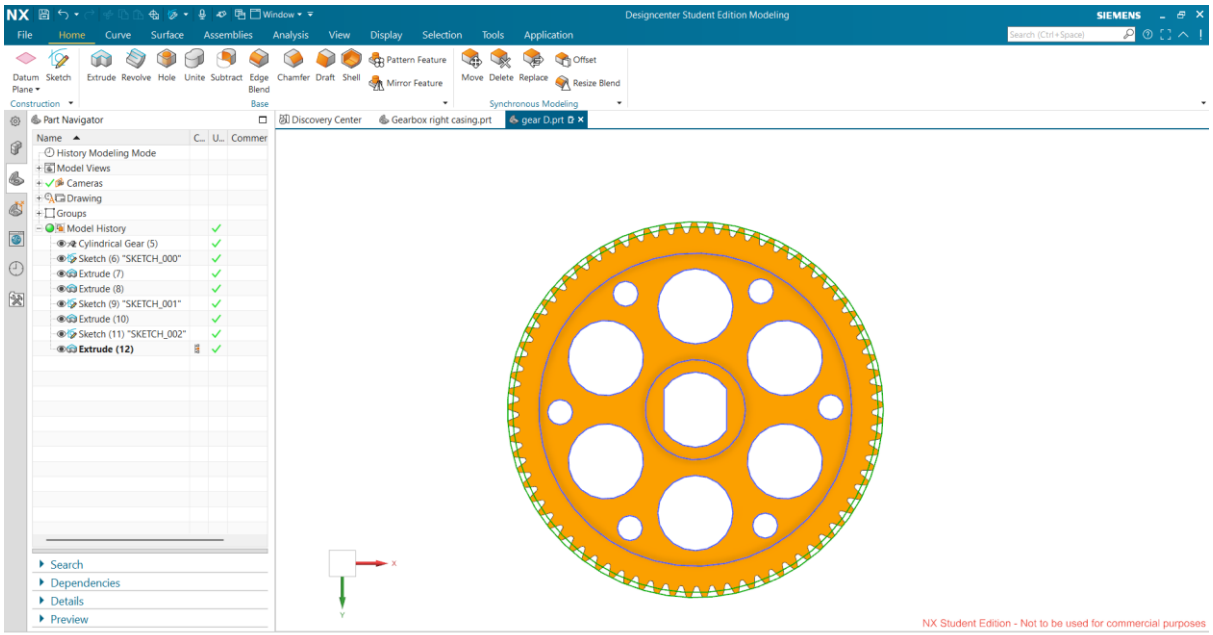
Step 1: Create a sketch for casing of Gearbox



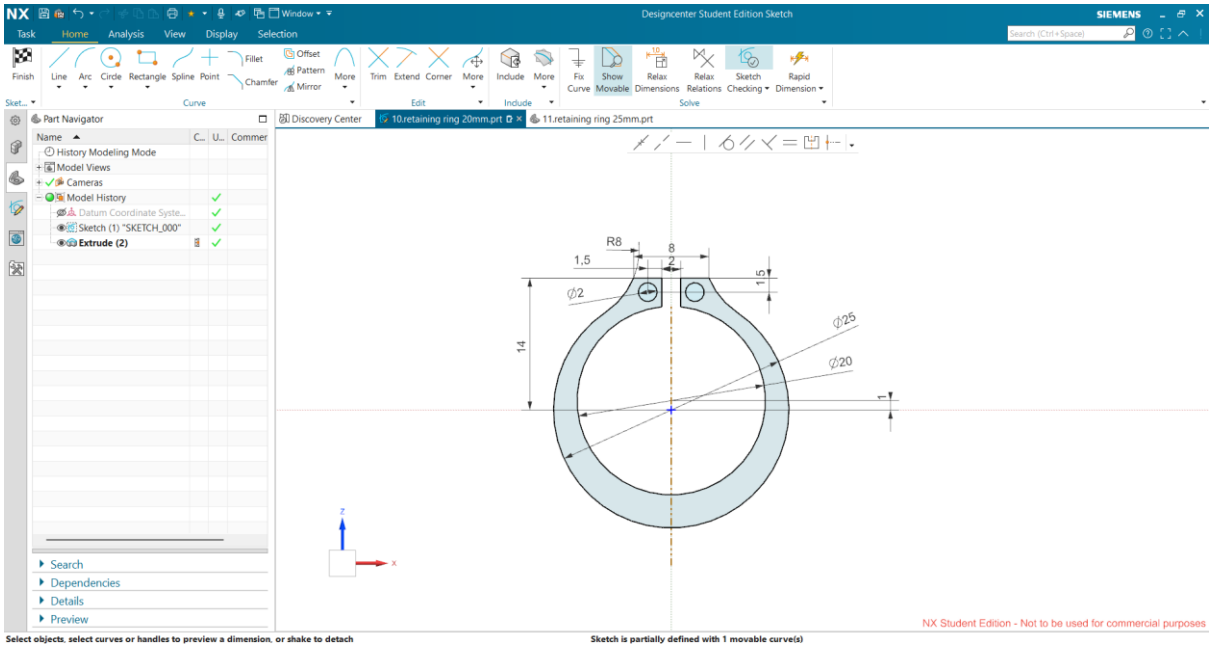
Step 3: Sketch the Gear A, B, C, D



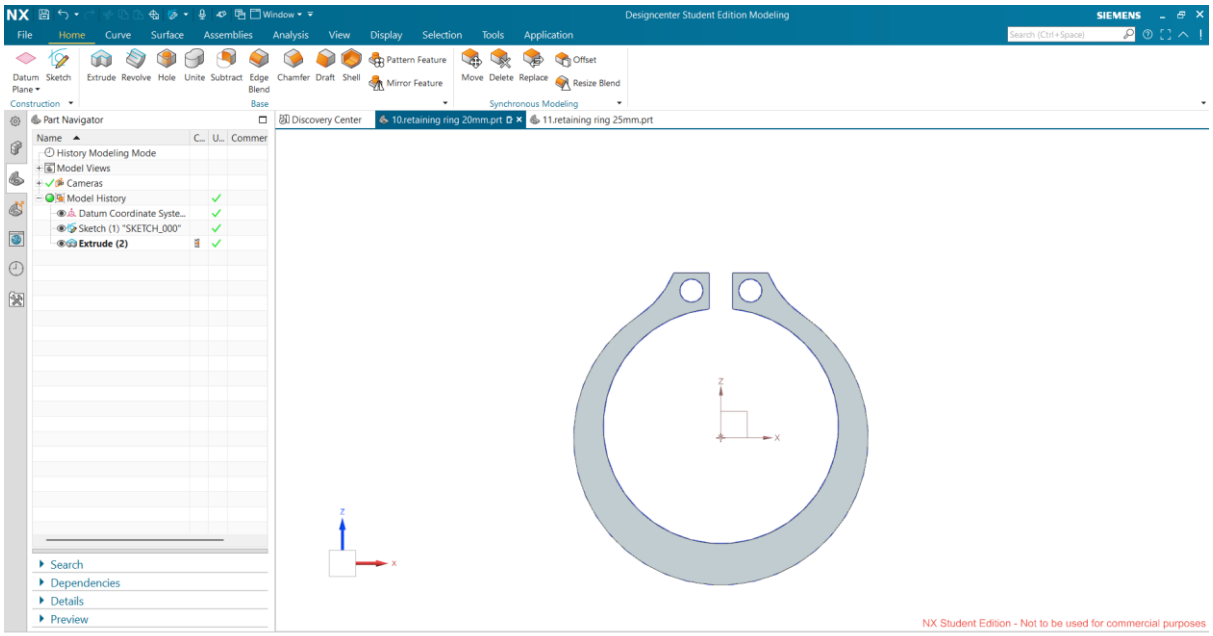
Step 4: Extrude the Gear A, B, C, D



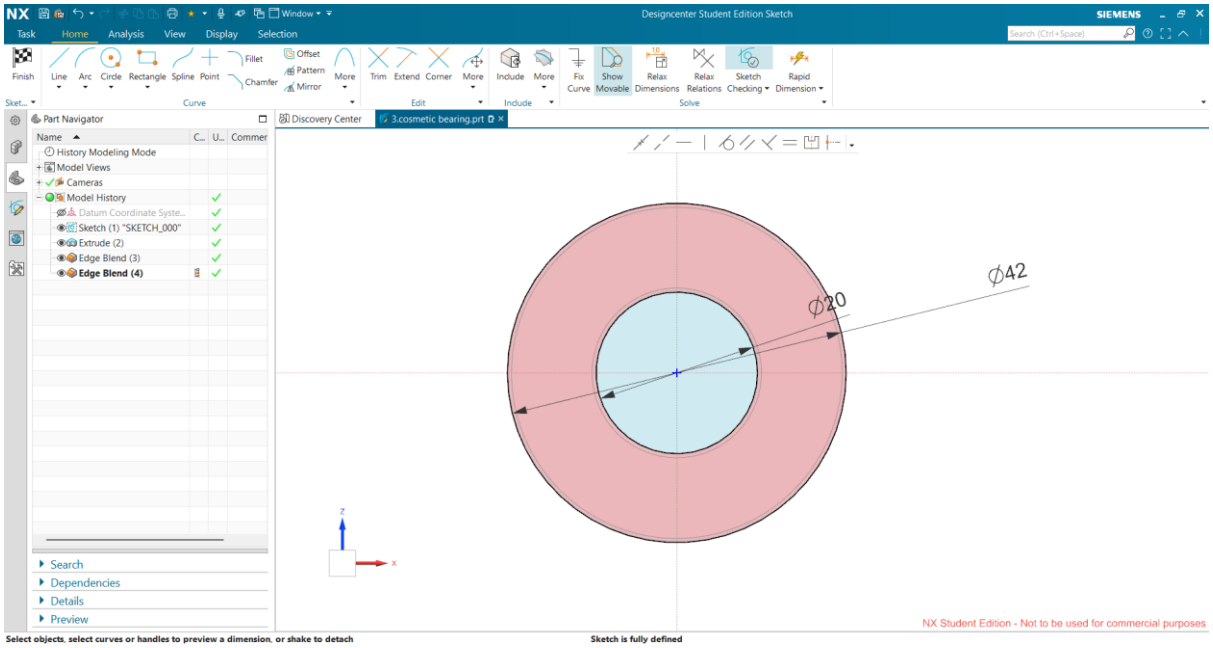
Step 5: Sketch the retaining ring for 20mm



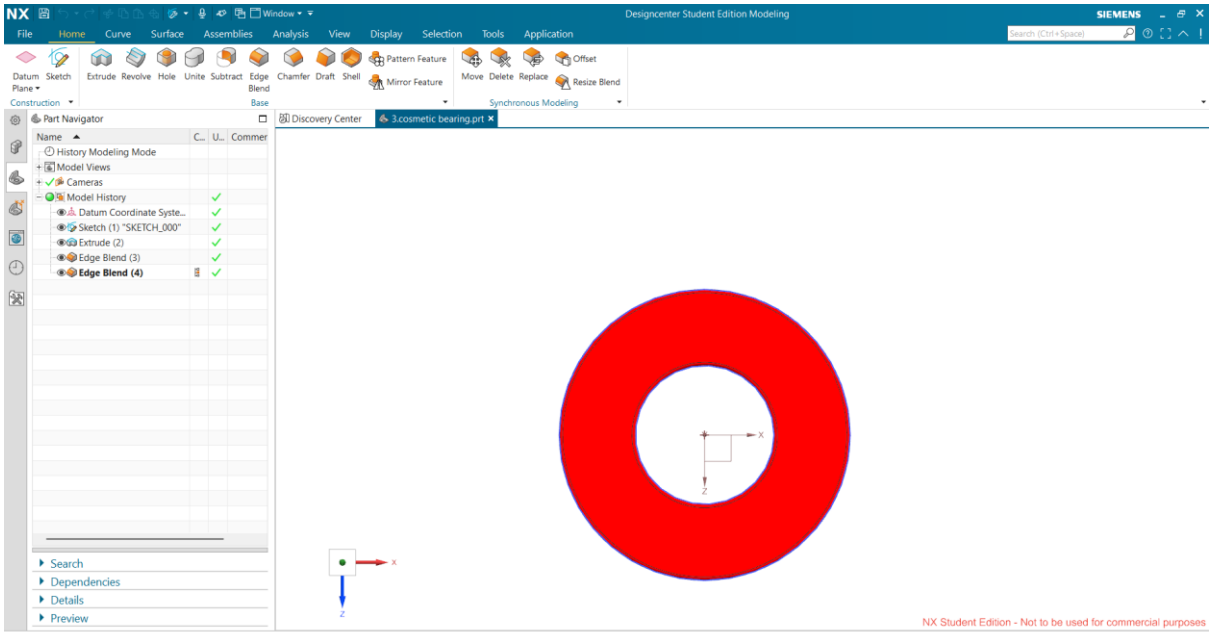
Step 6: Extrude the 20mm retaining ring



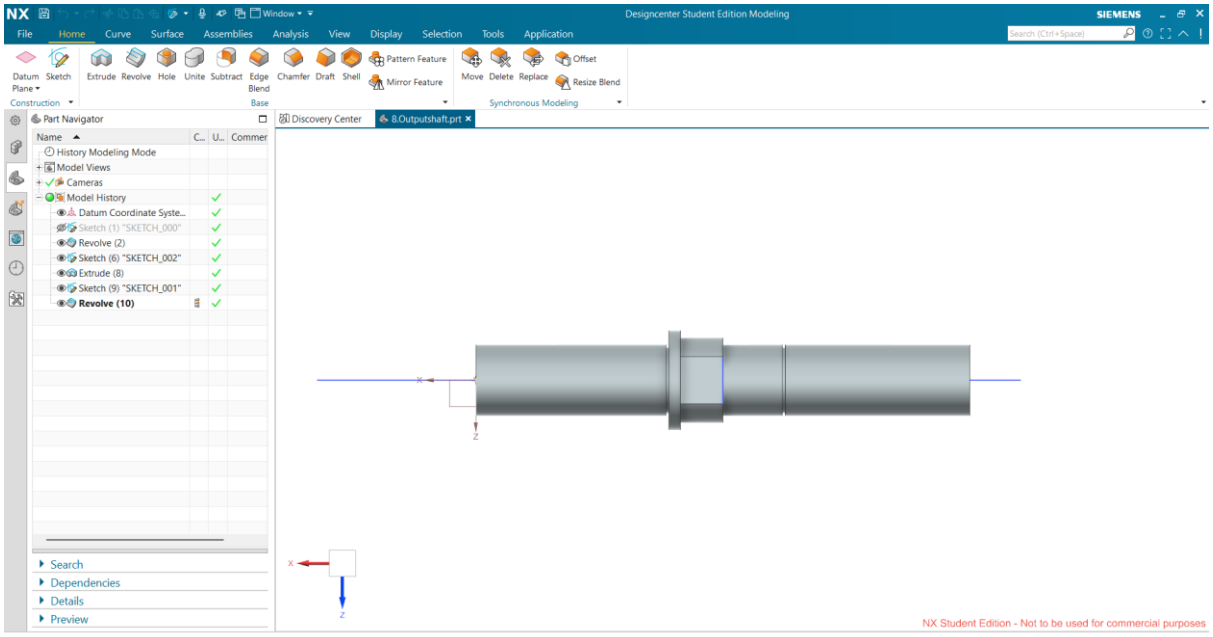
Step 7: Sketch the Cosmetic bearing A&B



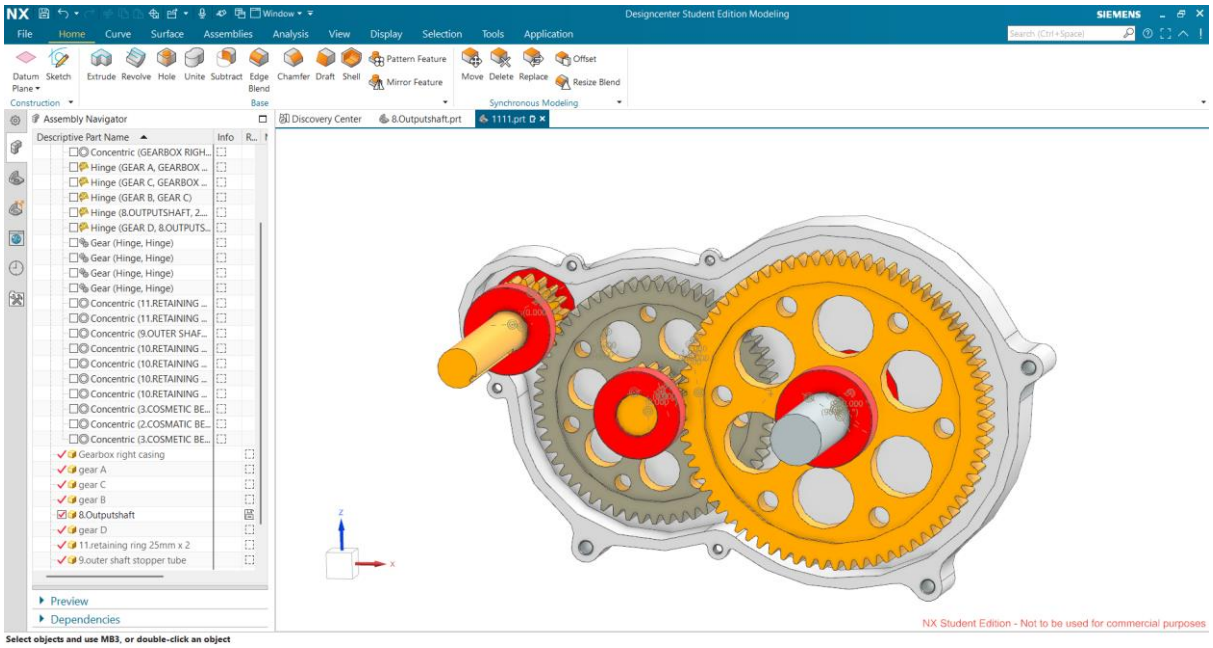
Step 8: Extrude the Cosmetic Bearing A&B



Step 9: Sketch and Extrude the Output Shaft



Step 10: Now Assemble Drawn Parts using Constrains



CHAPTER 5 – IMPLEMENTATION

Sketch:

1. Starting a New Sketch

Open NX CAD: Launch the software and open your project or start a new one. Create a New Sketch: Navigate to the "Home" tab and select “Sketch” from the "Direct Sketch" group. Select the Sketch Plane: Choose the plane 6(XY, YZ, or ZX) or a planar face of an existing part where you want to draw the sketch.

2. Sketching Tools and Constraints

Sketch Tools: Use the tools available in the "Sketch Task Environment" such as lines, circles, arcs, rectangles, polygons, splines, and ellipses to draw your desired shape.

Constraints: Apply geometric constraints (coincident, parallel, perpendicular, tangent, etc.) and dimensional constraints (length, angle, radius, diameter, etc.) to control the geometry precisely. Constraints help in maintaining relationships between different sketch entities.

3. Dimensioning

Adding Dimensions: Use the "Dimension" tool to add dimensions to your sketch. Dimensions define the size and position of sketch entities. Editing Dimensions: Double-click on a dimension to edit its value and update the sketch accordingly.

4. Modifying the Sketch

Modify Tools: Use modify tools like trim, extend, offset, fillet

chamfer, and mirror to adjust the sketch entities.

Move and Rotate: You can move or rotate sketch entities to fit your

design requirements.

5. Sketch Analysis

Check for Errors: Use the "Sketch Solver" to check for over constrained, under-constrained, or conflicting constraints.

Fix Issues: Resolve any issues highlighted by the Sketch Solver to ensure the sketch is properly constrained.

Finish Sketch: Once the sketch is complete, click "Finish Sketch" to exit the sketch environment. The sketch can now be used to create 3D features.

Extrude

6. Exiting the Sketch

1. Starting the Extrude Command

Select Extrude Tool: Go to the "Home" tab, and in the "Feature" group, click on "Extrude."

Select Sketch: If a sketch is not already selected, click on the sketch you want to extrude.

2. Setting Extrude Parameters

Direction and Distance:

Direction: Choose the direction in which you want to extrude. By default, the extrusion direction is perpendicular to the sketch plane. You can reverse the direction or specify a custom vector.

Distance: Enter the distance for the extrusion. This can be a specific value, or you can use options like "Until Next," "Until Selected," "Through All," etc.

Boolean Operations: Decide the type of operation:

Create New: Creates a new solid body.

Subtract: Removes material from an existing body (for creating cuts or holes).

Intersect: Keeps only the intersecting volume between the extruded shape and the existing body.

Limit Options: You can specify different start and end limits for the extrusion, such as:
Symmetric Value: Extrudes equally on both sides of the sketch plane.

3. Advanced Options

Draft Angle: Apply a draft angle to the revolution for tapering the sides.

Offset: Offset the revolution start or end position from the sketch plane.

Thin Feature: Create thin-walled features by specifying a wall thickness for the revolution.

4. Preview and Confirm

Preview: Use the preview option to visualize the revolution before finalizing it. This helps in checking if the revolution meets your design requirements.

OK/Apply: Click "OK" or "Apply" to finalize the revolution.

5. Editing a Revolve Feature

Edit Revolve: if you need to modify the revolution, right-click the revolve feature in the feature tree and select "Edit Feature." Make the necessary changes and confirm.

Edge blend

1. **Starting the Edge Blend Command**
   Select Edge Blend Tool: Go to the "Home" tab, and in the

"Feature" group, click on "Edge Blend."

2. Selecting Edges for Blending

Select Edges: Click on the edges you want to blend. You can

return bd select multiple edges to apply the same blend radius.

Blend Radius:

Specify Radius: Enter the radius for the blend. This determines how rounded the edge will be.

Multiple Radii: You can apply different radii to different edges by selecting them individually and specifying the radius for each.

Propagation Options:

Edge: Blends the selected edge only.

Face: Propagates the blend around the faces adjacent to the

selected edge.

Tangent: Extends the blend to tangent continuous edges.

Feature: Applies the blend to all edges of a selected feature.

4. Advanced Options

Variable Radius: Create a variable radius blend where the radius

changes along the length of the edge.

Symmetric: Apply a symmetric blend on both sides of the edge.

5. Preview and Confirm

Preview: Use the preview option to visualize the blend before finalizing it. This helps in checking if the blend meets your design requirements.

OK/Apply: Click "OK" or "Apply" to finalize the edge blend.

6. Editing an Edge Blend Feature

Edit Edge Blend: If you need to modify the blend, right-click the edge blend feature in the feature tree and select "Edit Feature." Make the necessary changes and confirm.

Chamfer

1. Starting the Chamfer Command

Select Chamfer Tool: Go to the "Home" tab, and in the "Feature" group, click on "Chamfer."

2. Selecting Edges for Chamfering

Select Edges: Click on the edges you want to chamfer. You can select multiple edges to apply the same chamfer parameters.

3. Setting Chamfer Parameters

Chamfer Type:

Symmetric: Creates a chamfer with equal distances on both sides of the

edge. You specify a single distance.

Angle: If using the angle type, specify the distance and the angle for the chamfer.

4. Advanced Options

Propagation Options: Choose how the chamfer propagates around the selected edges:

Edge: Chamfers only the selected edge.

Face: Propagates the chamfer around the faces adjacent to the selected edge. Tangent: Extends the chamfer to tangent continuous edges.

Feature: Applies the chamfer to all edges of a selected feature.

5. Preview and Confirm

Preview: Use the preview option to visualize the chamfer before finalizing it. This helps in checking if the chamfer meets your design requirements.

OK/Apply: Click "OK" or "Apply" to finalize the chamfer.

6. Editing a Chamber Feature

Edit Chamfer: If you need to modify the chamfer, right-click the chamfer feature in the feature tree and select "Edit Feature." Make the necessary changes and confirm.

RESULT

The Two-Stage Reduction Gearbox was successfully designed and assembled using Siemens NX CAD. All components, including the spur gears, shafts, bearings, keys, retaining rings, and gearbox housing, were accurately modeled and assembled without interference. The completed assembly demonstrated proper gear meshing, correct shaft alignment, and smooth transmission of power through two stages of gear reduction. The gearbox achieved the intended objective of reducing input speed while increasing output torque. The project also verified the effectiveness of Siemens NX CAD in developing precise mechanical assemblies and enhanced practical knowledge of gearbox design, assembly techniques, and mechanical power transmission. The final model is suitable for further analysis, simulation, and manufacturing applications.

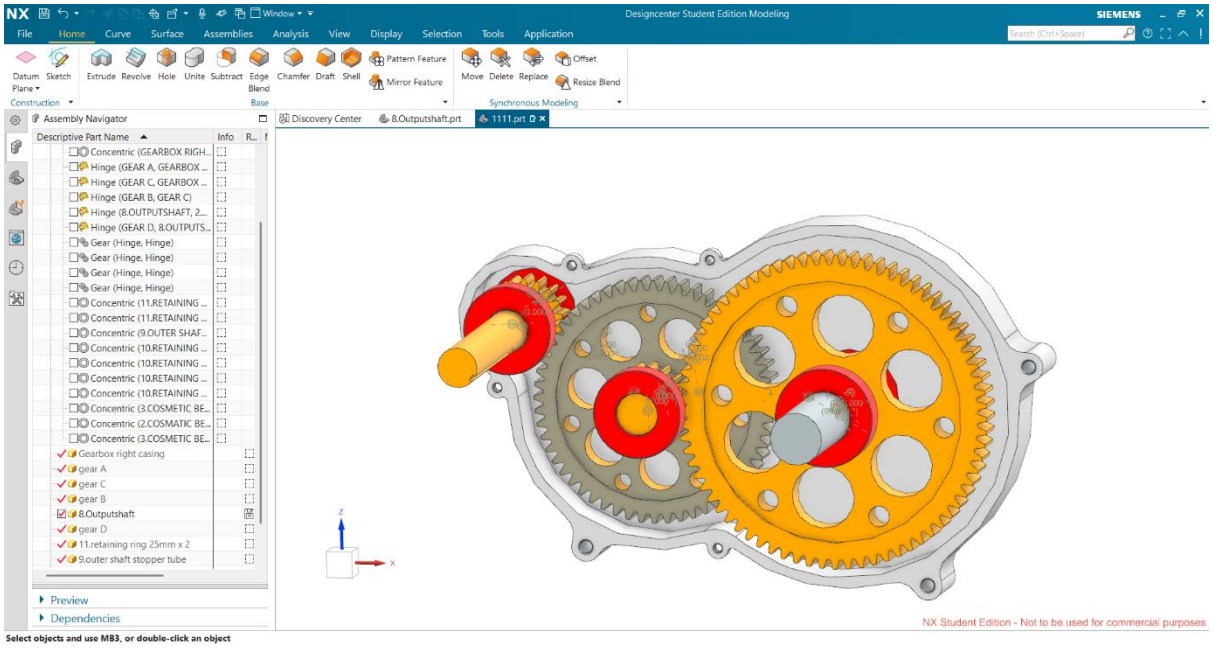
CHAPTER 6 – CONCLUSION AND FUTURE WORK

Conclusion

The design and assembly of the **Two-Stage Reduction Gearbox using Siemens NX CAD** was successfully completed. All components were accurately modelled and assembled to demonstrate proper gear meshing, shaft alignment, and smooth power transmission. The project enhanced knowledge of gearbox design, mechanical power transmission, and CAD modelling while improving practical skills in assembly design using Siemens NX CAD. It also provided valuable experience in developing a functional mechanical system suitable for industrial applications.

Future Work

The gearbox model can be further improved by performing **motion simulation** and **Finite Element Analysis (FEA)** to evaluate its performance under different operating conditions. Future enhancements may include the use of **helical gears** for smoother operation, optimization of component dimensions to reduce weight, and prototype manufacturing using CNC machining or 3D printing. These improvements will increase the gearbox's efficiency, reliability, and industrial applicability.



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